

**TREATMENT AND EMERGENCY RECOMMENDATIONS FOR
Mitochondrial Acetoacetyl Co-A Lyase Deficiency
(also called Beta-Ketothiolase Deficiency and 3-Oxothiolase Deficiency)**

RE:
DOB:

This patient has Mitochondrial Acetoacetyl Co-A Lyase Deficiency (Mito AcAcCoA thiolase Deficiency) and is at risk for serious complications from this condition.

**Please see last section of this letter for
Emergency Treatment during Decompensation**

Definition: MitoAcAcCoA thiolase deficiency is an inherited disorder of amino acid metabolism. Persons with this condition cannot fully metabolize the essential amino acid Isoleucine. There is the potential for a significant ketosis and acidosis if the person becomes catabolic.

Signs of Decompensation: Vomiting, lethargy, ketoacidosis, and hyperglycemia may mimic diabetic ketoacidosis.

Treatments:

- A. ***Prevention of decompensation:***
 - 1. A diet low in ISOLEUCINE
 - 2. Avoiding long periods of fasting
 - 3. Prevention of catabolism by providing carbohydrate and fluids.
- B. ***Treatments during Ketoacidosis:***
 - 1. Provide IV Fluid resuscitation with D10 / ½ normal saline at maintenance and keep blood glucose at high normal (120mg/dL) levels.
 - 2. Administer L-carnitine loading dose of 100mg/kg IV push over 60 minutes then 50mg/kg IV every 6 hours.

Please see next page...

3. Bicarbonate is not necessary unless pH is less than 7.1 (dose 1meq/kg/IV/hour).
4. Laboratory Studies
 - i. Metabolic Profile (electrolytes and glucose)
 - ii. NH₃
 - iii. Blood Gasses
5. Treat underlying cause of illness
6. **Notify the metabolic specialist immediately (after business hours this number allows you to speak with the geneticist on call:**

Kentucky Childrens Hospital
Division of Genetics and Metabolism

R. Stephen Amato, MD, PhD

859-323-0396

Kentucky Clinic – J420

740 S. Limestone Street

Lexington, KY. 40536-0284

(After Hours: Please call 1-800-888-5533)